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Unit – 1 : Session – 10 : SLO – 2 | NAND gate and universal implementation

■ Part B – Multiple Choice Questions

1. B — $(A \cdot B)'$
2. B — It can implement AND, OR, NOT
3. B — Connect both inputs together
4. B — $(A' \cdot B)'$
5. D — All of the above

■ Part A – Match the Following

1. NAND gate — c) Output is 0 only when all inputs are 1
2. NAND as NOT — a) Connect both inputs together
3. NAND as AND — e) Invert NAND output $\rightarrow (A \cdot B)'' = A \cdot B$
4. NAND as OR — b) De Morgan: $A + B = (A' \cdot B)'$
5. Universal Gate — d) Any logic function can be built using it

■ Part C – Activity Questions

1. 2-input NAND truth table: $00 \rightarrow 1, 01 \rightarrow 1, 10 \rightarrow 1, 11 \rightarrow 0$.
2. NAND as NOT: tie both inputs together $\rightarrow Y = (A \cdot A)' = A'$. Truth table: $0 \rightarrow 1, 1 \rightarrow 0$.
3. OR using only NAND gates: invert A $\rightarrow A'$, invert B $\rightarrow B'$ (two NANDs as NOTs), then NAND the results: $(A' \cdot B)' = A + B$. Total 3 NAND gates.